



Transportation's Systems 20-5751

PROJECT PART 4: MULTI-DIMENSIONAL OPTIMIZATION USING GRADIENT DESCENT

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1 Introduction

In this part, you will extend your optimization knowledge to multi-variable problems using the Gradient Descent algorithm.

This problem represents a simplified version of flow distribution across parallel routes in a transportation network.

1.1 Flow Optimization on Parallel Links

Consider 4 parallel links connecting the same origin and destination. The total demand is:

$$D$$

Flows must satisfy:

$$x_1 + x_2 + x_3 + x_4 = D$$

Each link follows the BPR travel time function:

$$T_i(x_i) = T_i^0 \left(1 + B \left(\frac{x_i}{C} \right)^4 \right)$$

The total system cost is:

$$Z(x_1, x_2, x_3, x_4) = \sum_{i=1}^4 x_i \cdot T_i(x_i)$$

Instructions:

1. Implement **Gradient Descent**
2. Minimize $Z(x_1, x_2, x_3, x_4)$
3. Enforce the constraint:

$$x_1 + x_2 + x_3 + x_4 = D$$

Hint: You may:

- Use a penalty function:

$$Z'(x) = Z(x) + \mu(x_1 + x_2 + x_3 + x_4 - D)^2$$

Student-Specific Parameters:

Let ID be your student ID:

- $seed = \text{int}(\text{hashlib.sha256}(\text{student_id.encode()}).\text{hexdigest}(), 16) \% 1e8$
- $D = 10000 + (\text{int}(seed) \% 10000)$
- $B_1 = 0.1 + \frac{\text{int}(seed \% 10)}{100}$
- $B_2 = 0.2 + \frac{\text{int}(seed \% 10)}{100}$
- $B_3 = 0.3 + \frac{\text{int}(seed \% 10)}{100}$
- $B_4 = 0.4 + \frac{\text{int}(seed \% 10)}{100}$
- $T_1^0 = 5 + (\text{int}(seed) \% 5)$
- $T_2^0 = 7 + (\text{int}(seed) \% 7)$
- $T_3^0 = 9 + (\text{int}(seed) \% 9)$
- $T_4^0 = 11 + (\text{int}(seed) \% 11)$

Stopping Criteria:

You choose the best criteria for the problem.

Final Output (IMPORTANT NUMBER):

Let $x_1^*, x_2^*, x_3^*, x_4^*$ be the optimal flows.

Compute:

$$R = \text{hash}(\text{tuple} \left(\lfloor \frac{x_1^*}{10} \rfloor, \lfloor \frac{x_2^*}{10} \rfloor, \lfloor \frac{x_3^*}{10} \rfloor, \lfloor \frac{x_4^*}{10} \rfloor \right)) \bmod 100000$$

1.2 Submission for Part 4

Submit:

- Your Gradient Descent implementation
- A report including:
 - Step size strategy
 - Convergence behavior
 - Final solution
- IMPORTANT NUMBER:
 - R